SET 8 – MATHEMATICS – IDENTITIES & SUMS
ANSWER KEY & MARKING SCHEME

Total Marks: 15 | Model Answers & Step-by-Step Breakdown

Q1. Cube Identity (2 Marks)

MODEL ANSWER

Evaluate 101^3 using an identity.

$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

$(100 + 1)^3 = 1000000 + 30000 + 300 + 1 = 1030301$

Answer: 1030301

KEYWORD MARKING SCHEME

Marks	Awarded for
1.0	Correct identity and splitting 101 = 100 + 1.
1.0	Correct expansion and answer 1030301.

Q2. Sum of Squares of Two Numbers (2 Marks)

MODEL ANSWER

If a + b = 5 and ab = 6, find $a^2 + b^2$.

$a^2 + b^2 = (a + b)^2 - 2ab$

$= 25 - 12 = 13$

Answer: 13

KEYWORD MARKING SCHEME

Marks	Awarded for
1.0	Correct identity.
1.0	Correct substitution and answer 13.

Q3. Sum of Cubes of Two Numbers (3 Marks)

MODEL ANSWER

If a + b = 5 and ab = 6, find $a^3 + b^3$.

$a^3 + b^3 = (a + b)^3 - 3ab(a + b)$

$= 125 - 3 \times 6 \times 5 = 125 - 90 = 35$

Answer: 35 (check: a = 2, b = 3 gives 8 + 27 = 35)

KEYWORD MARKING SCHEME

Marks	Awarded for
1.0	Correct identity.
0.5	Correct substitution.
1.5	Correct calculation and answer 35.

Q4. Sum of Cubes of Even Numbers (3 Marks)

MODEL ANSWER

Find $2^3 + 4^3 + 6^3 + \ldots + 16^3$.

$= 2^3(1^3 + 2^3 + \ldots + 8^3) = 8 \times \Sigma n^3$

$\Sigma n^3 = (8 \times 9 / 2)^2 = 36^2 = 1296$

$Sum = 8 \times 1296 = 10368$

Answer: 10368

KEYWORD MARKING SCHEME

Marks	Awarded for
1.0	Taking $2^3 = 8$ common.
1.0	Correct $\Sigma n^3 = 1296$ for n = 8.
1.0	Final answer 10368.

Q5. Proof by Induction – Sum of Squares (5 Marks)

MODEL ANSWER

Prove: $1^2 + 2^2 + \ldots + n^2 = n(n + 1)(2n + 1)/6$.

Step 1 (n = 1): LHS = 1, RHS = $1 \times 2 \times 3 / 6 = 1$. True.

Step 2: Assume true for n = k:

$1^2 + \ldots + k^2 = k(k + 1)(2k + 1)/6$

Step 3 (n = k + 1): Add $(k + 1)^2$:

$= (k + 1)[k(2k + 1) + 6(k + 1)] / 6$

$= (k + 1)(2k^2 + 7k + 6) / 6$

$= (k + 1)(k + 2)(2k + 3) / 6$

This is the formula with n = k + 1.

By mathematical induction, the result is true for all n.

KEYWORD MARKING SCHEME

Marks	Awarded for
1.0	Verifying the base case.
1.0	Stating the assumption for n = k.
2.0	Adding $(k + 1)^2$ and factorising correctly.
1.0	Concluding by induction.